LeWM reproduction

TwoRoom, PushT and Cube met our practical baseline check. Reacher fell short.

Environment	Paper	Primary	Repeats
TwoRoom	87%	90% (45/50)	74% / 90%
Reacher	86%	64% (32/50)	68% / 86%
PushT	96%	96% (48/50)	88% / 88%
OGBench Cube	74%	72% (36/50)	84% / 64%

Primary: seed 42. Repeats: seeds 43 and 44. Each score uses 50 cases. The predeclared check allowed a primary score up to 10 percentage points below the paper. This is a practical tolerance, not proof of equivalence.

What we ran. Frozen released LeWM checkpoints with the original Cross-Entropy Method (CEM) planner. Goals were 25 recorded actions ahead; the controller had 50 actions to reach them. All **15 panels passed the evidence audit**: 750 episodes and 37,500 controller actions, including the separately planned CEM sensitivity checks.

These are nearby-goal results from shared checkpoints. They do not establish safety, long-horizon performance or replication across training seeds.

What this means for our testing

- **Start from the tested baselines.** TwoRoom, PushT and Cube provide starting points for adapter studies. Reacher's shortfall needs diagnosis before gains there can be interpreted confidently.
- **Measure safe completion.** Compare adapters with native LeWM on matched cases and budgets, across multiple seeds. Report safe completion, violations and unfinished episodes together; test longer horizons separately.

Experiment completed 4 November 2026. Summary revised 5 November 2026.

Full results, uncertainty and runtime caveats | Videos by environment

AP reproduction

AP completed 306/512 cases; matched final-goal CEM completed 47/512.

Environment	Our AP	Our CEM	Paper AP
TwoRoom	58/128	2/128	60/128
PushT	88/128	4/128	89/128
Reacher	100/128	32/128	97/128
OGBench Cube	60/128	9/128	60/128

Each score covers 128 original cases. AP met our predeclared practical check in **4 of 4 environments**: no more than 10 percentage points below the paper. This is not proof of statistical equivalence.

What changed. Anchored Planning (AP) retrieves nearby targets from recorded experience to guide the frozen world model toward a distant goal. Its matched control uses the same checkpoint and CEM optimizer, scoring plans directly against the final goal. No model was trained.

A longer-range test. Goals lie 100 to 150 recorded actions ahead, compared with 25 in the earlier baseline; their temporal offset is supplied. The full comparison contains 1,024 episodes. These are fixed recorded-goal tasks; they do not establish unrestricted task completion, safety or robustness across training seeds. The nearby-goal baseline is a separate, unpaired comparison.

What this means for our testing

- **Use AP as the longer-range reference.** Test adapters against AP and matched final-goal CEM, keeping cases, available information and interaction budgets comparable.
- **Test safety explicitly.** Add hazards and measure safe completion, violations and unfinished episodes on held-out cases. Task-success gains alone establish neither safety nor an evolutionary advantage.

Completed 5 November 2026. All 1,024 cases independently audited.

Full results, uncertainty and runtime caveats | Matched ghost comparisons